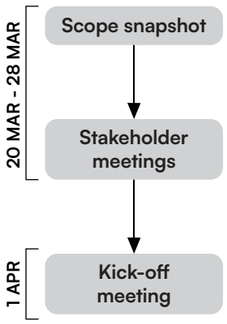
PROJECT MANAGEMENT PORTFOLIO

2024 F1 in Schools World Finals | Saudi Arabia

Initiation Process	Page 2
Project Schedule	Page 3
Budget & Resource Management	Page 4
Roles, Responsibilities & Communications	Page 5
Risk Management	Page 6
Monitoring & Controlling	Page 7

INITIATION PROCESS

INITIATION PROCESS



2.1: Flowchart of the initiation process we took following our National Final
Credit: Lunar

Scope snapshot

Project Manager (Leo) conducted an initial scope before discussing with stakeholders (forms basis for project charter).

- Key deliverables
- Objectives
- Communication strategy
- World Finals overview

Stakeholder meetings

Holding meetings with various stakeholders allowed us to hear the expectations of them before we held our official team kick-off meeting, as well as both professional and F1 in Schools tailored advice to ensure our project is successful.

Attendance
- Jamie Watson
- Thivi Athipar

F1 in Schools teachers

- Nick Potter
- Jenson Galvin

Professional mentors

Kick-off Meeting

During our kick-off meeting, as a team, we answered 6 key questions to define the goals and identify the project deliverables, of which were:

WHY is this project being initiated?

To develop skills that extend beyond traditional schooling, but to ultimately achieve our goal of becoming **WORLD CHAMPIONS.**

WHO are we completing this work for?

Ultimately ourselves with our work tailored towards the competition scorecards and judges (primary stakeholders). This work is also for all of our other stakeholders, identified in stakeholder register (see page 5).

WHAT are we going to deliver?

All associated deliverables with the F1 in Schools World Final, including 2 race cars, 3 portfolios, a 10 minute verbal presentation, pit display and brand identity (broken down further on WBS).

WHEN will we produce these deliverables?

Primarily from April 1 - August 17 (improvised deadline).

WHERE will the deliverables be produced?

Mostly at school, as well as remotely from home (hybrid).

HOW are we going to achieve our goals and measure our success?

By collaborating as a team and regularly monitoring our progress through meetings and routines in an agile manner. We will measure our progress through the quality and completion of deliverables, and evaluate our success from our final scores and rank at the World Finals.

To provide further clarity towards our project, we also discussed the following agenda items:

- Overview of the 2024 World Finals
- Recap of National Finals
- Introduction to Jira software
- Communication plan
- Preliminary budget
- Preliminary timeline

Project Charter

PROJECT TITLE	Lunar F1 in Schools 2024 World Finals
DUE DATE	17 August 2024 (improvised)
BUDGET	$65,000 (AUD)
TEAM MEMBERS	Leo Jansen (Project Manager), Oliver Chernikeeff, Carey Tong, Max Zylberman, Liam Jones
OBJECTIVES	**Vision:** Become World Champions *

Improvised deadline

Because this charter was created before we knew our competition date and location, the deadline we set was improvised, so we had a date to work towards.

PROJECT DESCRIPTION	The World Finals is the continuation of our F1 in Schools 2023/24 campaign, where we compete to achieve the best overall scores by completing all associated deliverables with the competition to the highest standard we can.
PROJECT JUSTIFICATION	• Become World Champions • Apply classroom knowledge • Foster soft skills to pursue careers in STEM • Engage in a thrilling competitive environment
KEY DELIVERABLES	• Race Cars x2 • Pit Display • Portfolios x3 • Verbal Presentation • Interviews x5 • Brand Identity
PREDOMINANT CONSTRAINTS	• Timeframe • Burnout • Procrastination • Budget ($65,000) • Team size (5 members) • School events (e.g exams) • School hours • Time away (e.g sickness)
PROJECT ASSUMPTIONS	• We will compete at the World Finals • All team members and roles will remain the same • We will meet our due date
KEY STAKEHOLDERS	• Judges • All team members • Teachers in Charge • Sponsors • Mentors • Brighton Grammar School
PREDOMINANT RISKS	• Missing submissions • Car breakages • Supplier delays • Lack of financial backing • Burnout • Scope Creep
KEY RESOURCES	• **Cash** (critical) • CNC machine • 2D/3D design software • Car materials • Pit display materials • Printed documents

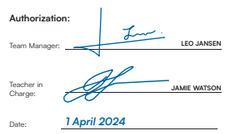
2.3: Our World Finals Project Charter
Credit: Lunar

Cash

Because of how critical cash is to attend this competition, it was highlighted to be immediately actioned upon initiation.

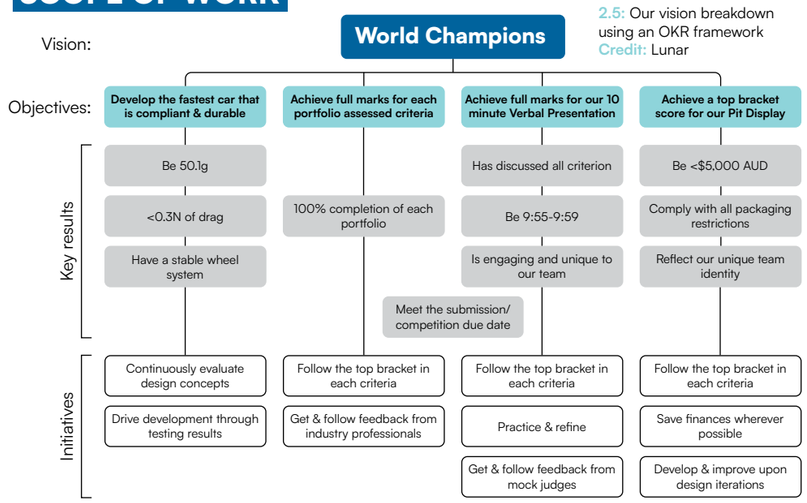
***Vision**

Our mission alongside our vision is to equip future F1 in Schools teams at our school with the knowledge to achieve success, mentoring them whilst we undertake our own project. This outcome is also what we aimed for even if we don't achieve our vision (1st place).



2.4: Signature of approval for project **GO-AHEAD!**
Credit: Lunar

SCOPE OF WORK



2.5: Our vision breakdown using an OKR framework
Credit: Lunar

Objectives & Key Results (OKR)

The expanded OKR framework below (figure 2.5) allowed us to have measurable goals for us to aim for, broken down with initiatives for us to follow, providing a further sense of relevance and motivation towards the project. This diagram also depicted how we could achieve high marks in every category, rather than best marks in one, and neglecting others, which will help us achieve a higher score.

Pit Display

Because the Trade Display is worth little points despite its high cost and effort, we concluded that full marks would be unfeasible to aim for. Thus our finances & time were directed more at other elements, however top bracket marks were still aimed for.

ACCEPTANCE CRITERIA

Testing and Assessment

Scorecard criterion

For any deliverable score, 1 of the 5 scorecards must be reviewed using the highest bracket, and examined closely to ensure all areas on the scorecard have been completed to achieve full marks.

Cars

All cars must adhere to:

- Every regulation and specification
- Hit our key results (figure 2.5) - unless not possible
- Be closely examined for any cracks/defects

Acceptance

Because not all deliverables need to be treated as strictly, we categorised each deliverable on our work breakdown structure as either:

- **High risk:** if the result of an error loses substantial points.
- **High expense:** if completing it costs more than $50 in total.
- Or both or neither.

If high risk OR high expense

The approval of at least 2 relevant team members and a teacher in charge is needed for a task to be signed off - 1 being Leo (Team Manager), and another being Oliver (Engineering Director) if it is an Engineering activity.

If BOTH high risk and high expense

The approval of all team members and a teacher in charge is needed for a task to be signed off. This collective review helps catch errors and oversights, ensuring more accurate and reliable outcomes.

If neither

The task can just be classified as "done" on our project management software by the team member completing the task.

Note that the involvement of specific team members in the approval process does not imply the exclusion of others, but rather highlights the roles necessary for final approvals.

SCOPE STATEMENT

Project Overview:

TITLE	Lunar F1 in Schools 2024 World Finals		
DUE DATE	17 August 2024	**PROJECT MANAGER**	Leo Jansen
VISION	**Become World Champions** • Develop the fastest car that is compliant & durable • Achieve full marks for each portfolio assessed criteria • Achieve full marks for Verbal Presentation • Achieve a top bracket score for Pit Display		
MISSION	**Help new teams at our school achieve success** • Provide mentorship & pass on our experience • Host introduction sessions • Buy assets that other teams can use		
DELIVERABLES	• Final cars x2 • Prototype cars • Pit Display • Engineering folio • PM folio • Enterprise folio • Verbal Presentation • Engineering interview • Pit display interview • PM interview • Enterprise interview • Brand Identity • Pit display brief • Sponsorship prospectus • Testing document • Statement of Work		

CONSTRAINTS	• Timeframe • Burnout • Procrastination • Budget • Team size (5 members) • School events (e.g exams) • School hours • Time away (e.g sickness)
BUDGET	**Total:** $65,000 AUD • **Cars** $4,000 • **Pit Display** $3,000 • **Uniform** $1,000 • **Registration** $20,000 • **Travel** $30,000 • **Contingency** $7,000

Project Scope:

IN SCOPE	• Research & development of our final car • Acquisition of funds ($65,000) • Acquisition of mentors • Submission of electronic elements • Presentation of interviews • Creation of pit display
OUT OF SCOPE	• Earning of profit • Unnecessary expenses • Personal projects & expenses • Unrelated research • Sourcing machinery • Preparation for our own future competitions

PROJECT SCHEDULE

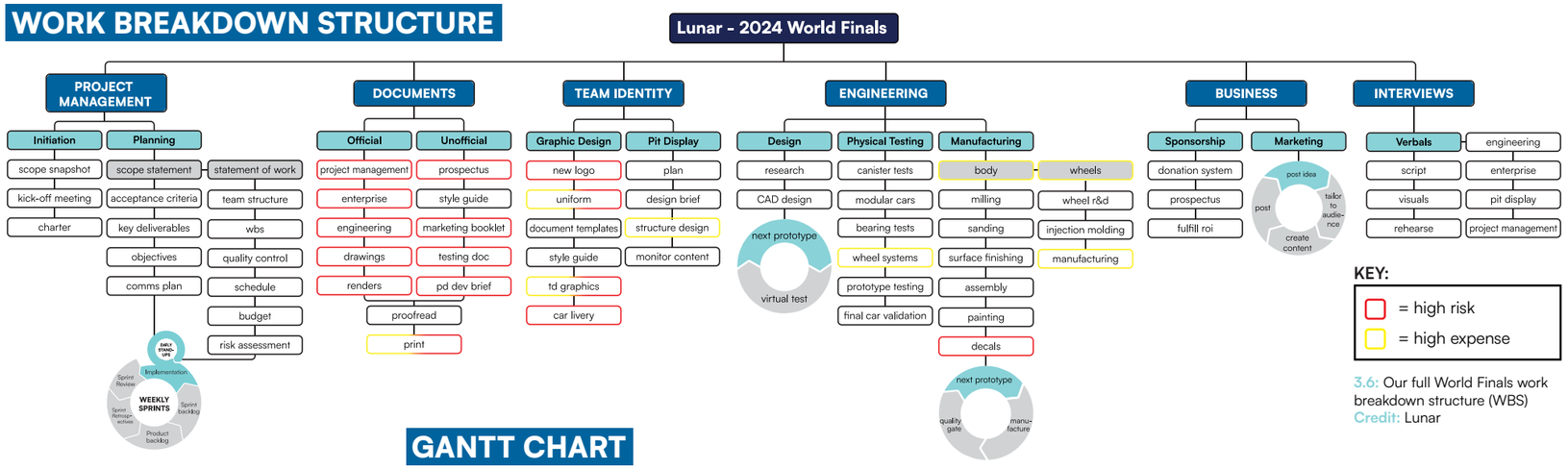
Execution Plan:

ACCEPTANCE CRITERIA	*see previous page
TEAM COLLAB-ORATION PLAN	**Daily standups:** During school lunchtime, before and after school with whoever is available. **Weekly meeting:** Thursday lunchtime - all members required (room BH G01). **Digital Tools:** • Discord = regular text communication • MS teams = virtual team meetings & file storage, meeting agendas/recordings • OneNote = contribute to written documents • Jira = task management
MILESTONES	• Project initiation (1 April) • Rebrand (3 May) • Pit display & uniform ordered (21 July) • Car design freeze (13 July) • Final Work Freeze (17 August) • Submissions (unconfirmed)

Note that some overlapping aspects of our project charter and scope statement such as our acceptance criteria and key milestones were excluded from our charter. This is to ensure both documents are succinct and don't repeat too much information. Our project charter is intended to define and authorise the project at launch, whilst our scope statement provides an execution plan for us to follow when planning and executing our project.

2.6: Our World Finals full Scope Statement (formatted to fit within this portfolio)
Credit: Lunar

WORK BREAKDOWN STRUCTURE

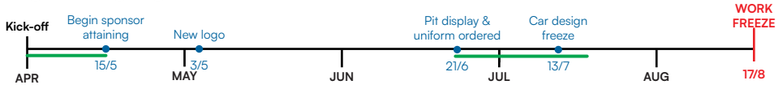


KEY:
- = high risk
- = high expense

3.6: Our full World Finals work breakdown structure (WBS)
Credit: Lunar

SCHEDULE

Work Timeline

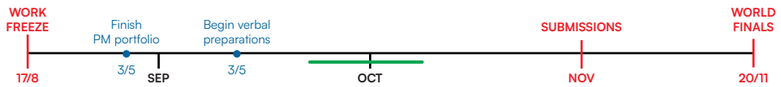


3.1: Our initial work timeline
Credit: Lunar

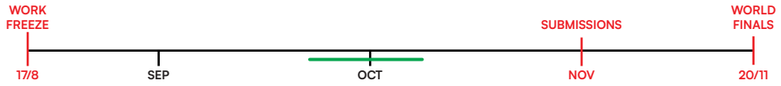


3.2: Our actual work timeline
Credit: Lunar

Refinement Timeline

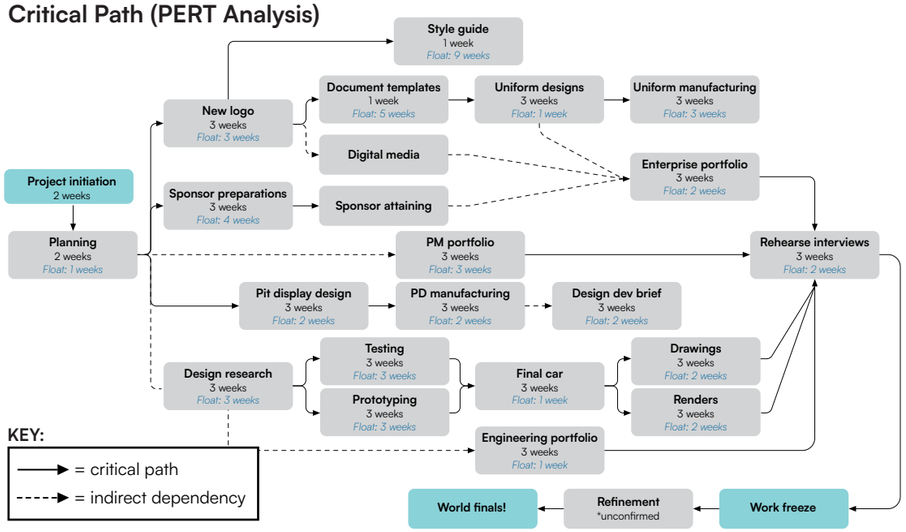


3.3: Our initial refinement timeline
Credit: Lunar



3.4: Our actual refinement timeline
Credit: Lunar

Critical Path (PERT Analysis)



KEY:
- = critical path
- = indirect dependency

3.5: Our World Finals PERT chart (configured to fit within this portfolio)
Credit: Lunar

Work vs Refinement

Between initiation and August 17, we conducted the majority of our work, which was presented in mock judging sessions for feedback. This was known as our work period. Following August 17 and our competition, we created a new schedule facilitating refinement and travel preparation. This was known as our refinement period. Breaking up our time management in this way ensured no elements were completed last minute, which has occurred in past competitions of ours.

Colour coding

Colour coded to specific team members for role clarity.
- Solid colour = planned timeframe
- Lighter colour = actual timeframe

Holidays & exams
- Holidays = more time to work on project, thus more tasks scheduled.
- Exams = less time for project (less tasks scheduled).

GANTT CHART

MONTH:			MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER
DAY:			18 25	01 08 15 22 29	06 13 20 27	03 10 17 24	01 08 15 22 29	05 12 19 26	02 09 16 23 30	07 14 21 28	04 11 08 25
NUMBER	ACTIVITY	DEPENDENCIES									
	PROJECT MANAGEMENT										
01	project initiation			initiation							
02	project charter/scope			project charter							
03	statement of work	01		sow							
	SPONSORSHIP										
04	fix donation page			donation page							
05	prospectus			prospectus							
	GRAPHIC DESIGN										
06	rebrand			rebrand							
07	document templates	06		document templates							
08	uniform	06		uniform design							
09	uniform manufacture	08		uniform manufacture (interim)							
10	style guide	06		style guide							
	PIT DISPLAY										
11	plan			plan							
12	structure design	11		structure design							
13	structure manufacturing	12		structure manufacturing							
14	graphics	12		graphics							
15	monitor content	12		monitor content							
16	design development brief	12		design development brief							
	DOCUMENTS										
17	pm portfolio			project management portfolio							
18	enterprise portfolio			enterprise portfolio							
19	engineering portfolio			engineering portfolio							
	INTERVIEWS										
20	verbal presentation						verbal presentation				
21	engineering	19					engineering interview				
22	trade display	16					enterprise interview				
23	enterprise	18					pit display interview				
24	project management	17					project management interview				

MONTH:			MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER
DAY:			18 25	01 08 15 22 29	06 13 20 27	03 10 17 24	01 08 15 22 29	05 12 19 26	02 09 16 23 30	07 14 21 28	04 11 08 25
NUMBER	ACTIVITY	DEPENDENCIES									
	RESEARCH										
01	engineering initiation			engineering initiation							
02	initial research			initial research							
03	canister tests	01		canister tests							
04	math model	01		math model	pit display structure						
05	bearing tests	04		bearing tests							
06	manufacturing equipment	01		manufacturing equipment							
07	paint tests	01		paint tests							
	WHEELS										
10	adjustable wheel system			adjustable wheel system							
11	research & development	02		research & development							
12	injection molding	06		injection molding							
	PROTOTYPING										
13	nose			nose							
14	canister chamber			canister							
15	sidepods			sidepods							
16	testing session 1	13-15		testing session 1							
17	rearpods			rear pods							
18	rear wing			rear wing							
19	front wing - interchangable			front wing							
21	testing session 2	17-19		testing session 2							
	FINAL CAR										
25	final car iterations	21					final iterations				
26	car livery	25					livery				
27	manufacture final	25					manufacture				
28	renders	27					renders				
29	quality check	27					quality check				
30	testing session 3	29					testing session 3				
31	drawings	30					drawings				

CAR DESIGN FREEZE | WORK FREEZE | ELECTRONIC SUBMISSIONS

KEY:
*date = School holidays | = Leo | = Oliver | = Carey | = External | → = Dependencies
*date = Exam period | = Max | = Liam | = Everyone

3.7: Our initial vs actual World Finals Gantt chart
Credit: Lunar

BUDGET & RESOURCE MANAGEMENT

RESOURCE IDENTIFICATION

Bill of Quantities

Item	Quantity	Unit Price	Total	Latest Date	Live Status	Primary Supplier
TRAVEL						
Registration			$10,000	10-Oct	Don't Have	F1 in Schools Ltd.
Flights			$15,000	6-Aug	Don't Have	Qatar Airways
Hotel			$15,000	6-Aug	Don't Have	Qatar Airways Holidays
Food			$6,000		N/A	
Incidental Expenses			$1,000		N/A	
		Total:	$47,000			
CAR						
Testing Equipment						
Track	1	$0	$0		Acquired	Already Have
Canisters	150	$0	$0		Acquired	Already Have
Force Transducer	1	$0	$0		Acquired	Already Have
Magnets	20	$1	$20	1-Jun	Don't Have	Magnets.com
		Subtotal:	20			
Car Components						
Foam Blocks	20	$0	$0		Acquired	REA Foundation
Balsa Blocks	30	$0	$0		Acquired	REA Foundation
3D Parts			$2,000	1-Jul	Don't Have	Objective 3D
Bearings	30	$0	$0		Acquired	Already Have
Wheels			$1,000	24-Jun	Don't Have	Waddell Engineering
Tether guides	20	$3	$50	20-May	Don't Have	Tacticals Brighton
		Subtotal:	$3,050			
Manufacturing Equipment						
CNC Machine	1	$0	$0		Acquired	Already Have
Dremel	2	$0	$0		Acquired	Already Have
Spray Paint	3	$50	$150	20-May	Don't Have	Bunnings Warehouse
Spray Putty	2	$30	$60	20-May	Don't Have	Bunnings Warehouse
Putty	1	$25	$25	20-May	Don't Have	Bunnings Warehouse
Primer Filler	1	$30	$30	20-May	Don't Have	Bunnings Warehouse
Clearcoat	1	$30	$30	20-May	Don't Have	Bunnings Warehouse
Waterslides	1	$33	$33	24-Jun	Don't Have	Print Gully
Sealant	1	$40	$40	20-May	Don't Have	Bunnings Warehouse
		Subtotal:	$368			
		Total:	$3,438			
PIT DISPLAY						
Table Structure	1	$1,500	$1,500	21-Jun	Don't Have	Spyder Displays
Backdrop Structure	1	$1,500	$1,500	21-Jun	Don't Have	Spyder Displays
Banners	3	$500	$1,500	21-Jun	Don't Have	Spyder Displays
Monitors	2	$0	$0		Acquired	Already Have
3D Logo	1	$100	$100	10-Aug	Don't Have	In-house
LED lights	1	$100	$100	10-Aug	Don't Have	Amazon
		Total:	$4,700			
UNIFORM						
Competition Polos	10	$35	$350	1-Sep	Don't Have	S-Trend Sportswear
Secondary Polos	8	$35	$280	21-Jun	Don't Have	S-Trend Sportswear
Jackets	5	$40	$200	1-Sep	Don't Have	Evo Sportswear
Trousers	5	$30	$150	21-Jun	Don't Have	Uniqlo
Shoes	5	$0	$0	21-Jun	Don't Have	Asics (Sponsorship)
		Total:	$980			
PRINTING						
Portfolios	9	$70	$630	20-Oct	Don't Have	Avenue Press
A4 Documents	8	$30	$240	1-Aug	Don't Have	Avenue Press
A5 Booklets	5	$15	$75	1-Aug	Don't Have	Officeworks
		Total:	$945			
CONTINGENCY & OTHER						
Contingency						
Project contingency fund			$7,000		N/A	
		Subtotal:	$7,000			
Software						
Fusion	5				Acquired	Autodesk
Keyshot	3	$0	$0		Acquired	Leap Australia Pty Ltd.
Ansys	2	$0	$0		Acquired	Leap Australia Pty Ltd.
Adobe Design Suite	5	$0	$0		Acquired	Adobe Inc.
Jira Software	5	$0	$0		Acquired	Atlassian
Squarespace	6	$15	$90	*Monthly	Acquired	Squarespace
		Subtotal:	$90			
Miscellaneous						
Australian Flag	1	$20	$20	15-Aug	Don't Have	Amazon
		Subtotal:	$20			
		Total:	$7,110			
TOTAL						
		TOTAL (AUD):	$64,173			
		TOTAL (SAR):	$158,507			

**Note all cost figures throughout this page are in AUD, unless otherwise specified*

Columns:

- **Item: WHERE** resources (cash) are used
- **Quantity**, **Unit Price**, **Total Price** — **Clear cash identification**
- **Date: WHEN** resources are to be acquired/ordered
- **Live status:** Tracking mechanism to refer back to
- **Primary supplier: HOW** resources are used

Blue totals:

The totals highlighted in blue, are estimates of total costs, because quantity and/or unit price cannot be identified.

This BOQ was created when the World Finals date and location were confirmed, so our travel costs could remain accurate.

4.1: Our full World Finals Bill of Quantities (BOQ) as at May 9th
Credit: Lunar

Budget

Travel	$47,000
Cars	$4,000
Pit Display	$5,000
Uniform	$1,000
Printing	$1,000
Contingency & Other	$7,000
Total:	**$65,000**

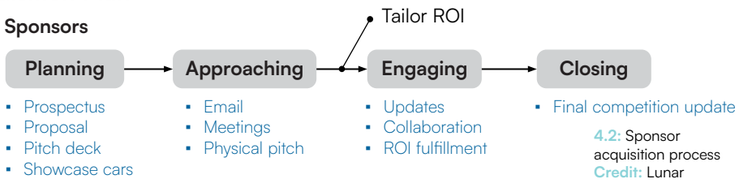
This budget was determined by rounding up the totals of each category in our BOQ, adding further budget padding on top of our contingency fund (12% of full budget). This is what was tracked and adjusted throughout the project.

4.2: Our World Finals inital budget (May)
Credit: Lunar

RESOURCE PLANNING

Acquisition Plan

Sponsors



4.2: Sponsor acquisition process
Credit: Lunar

Outsourcing supplies

Team member responsible

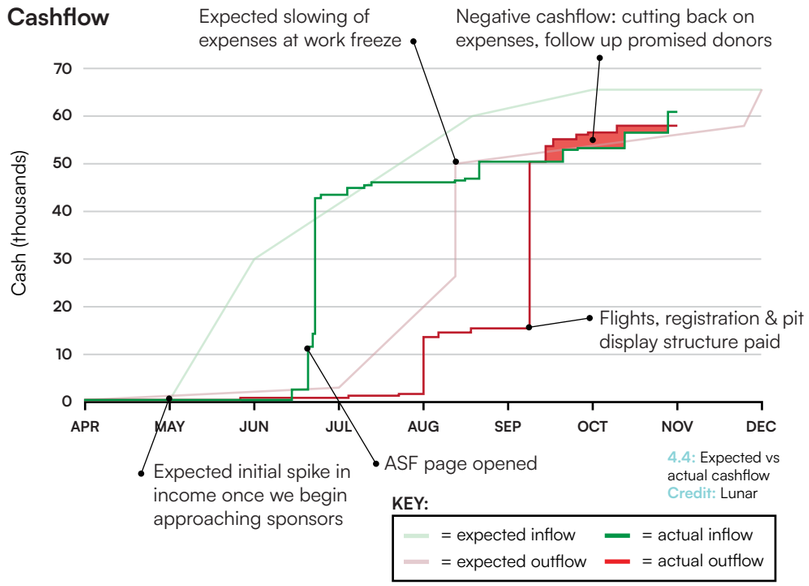
ITEM		PRIMARY SUPPLIER	SECONDARY SUPPLIER	SECONDARY SUPPLIER
3D printing	Liam	Objective 3D	ANCA	Marand
Wheels	Oliver	Waddell Engineering	Objective 3D	ANCA
Bearings	Liam	**Currently owned*	CGB Precision	Boca Bearings
Uniform	Leo	Alpinestars	S-Trend	My Promo
Pit display	Max	Spyder Displays	Rebul	In-house
Portfolio printing	Leo	Avenue Press	Officeworks	In-house
CFD	Max	Leap Australia	Ansys International	Siemens

= Actual supplier

Lead time/quality

4.3: List of external suppliers and who is responsible for each
Credit: Lunar

Cashflow



4.4: Expected vs actual cashflow
Credit: Lunar

Timing of key purchases

ITEM	PLANNED DATE	ACTUAL DATE
1st 3D printing order	10 May	21 May
Secondary uniform order	21 June	6 August
Pit Display structure order	21 June	30 August
Flights & hotel booking	6 August	30 August
2nd 3D printing order	8 July	15 September
Primary uniform order	26 September	20 October
Final 3D printing order	1 October	1 November

New team member needing more uniforms

Design freeze delayed

Cash Inflow Streams

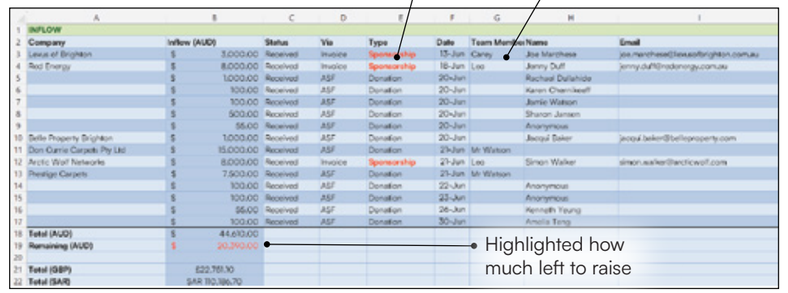
Australian Sports Foundation (ASF)	Invoices	Cash Sales
Tax-deductible donation page for supporters of our project to donate (1% surcharge).	Arranged business documents for sponsors to deposit sponsorship funding through school.	Sales from pop-up community stalls, merchandise & fundraising marketing activities.
Expected inflow: $13,500	$50,000	$1,500
Actual inflow: $38,941.68	$20,000	$0

TRACKING

Income and Expense Log

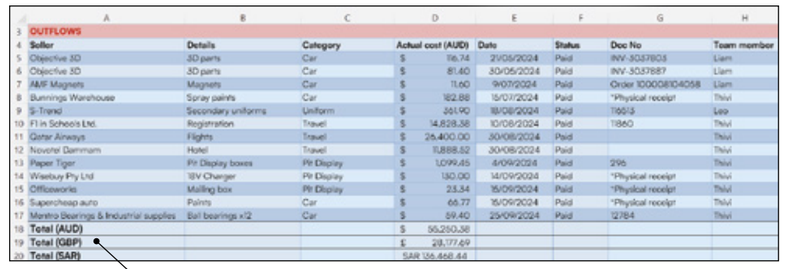
Sponsors highlighted from crowdfunding donors.

Team member handling sponsorship/expense



INFLOW								
Company	Inflow (AUD)	Status	Via	Type	Date	Team Member	Name	Email
Lexus of Brighton	$ 3,000.00	Received	Invoice	Sponsorship	13-Jun	Carey	Joe Marchese	joe.marchese@lexusofbrighton.com.au
Red Energy	$ 8,000.00	Received	Invoice	Sponsorship	18-Jun	Leo	Jenny Duff	jenny.duff@redenergy.com.au
	$ 1,000.00	Received	ASF	Donation	20-Jun		Rachael Dalkhide	
	$ 100.00	Received	ASF	Donation	20-Jun		Karen Chernikeeff	
	$ 100.00	Received	ASF	Donation	20-Jun		Jamie Watson	
	$ 500.00	Received	ASF	Donation	20-Jun		Sharon Jansen	
	$ 55.00	Received	ASF	Donation	20-Jun		Anonymous	
Belle Property Brighton	$ 1,000.00	Received	ASF	Donation	20-Jun		Jacqui Baker	jacqui.baker@belleproperty.com
Don Carrie Carpets Pty Ltd	$ 15,000.00	Received	ASF	Donation	21-Jun	Mr Watson		
Arctic Wolf Networks	$ 8,000.00	Received	Invoice	Sponsorship	21-Jun	Leo	Simon Walker	simon.walker@arcticwolf.com
Prestige Carpets	$ 7,900.00	Received	ASF	Donation	21-Jun	Mr Watson		
	$ 100.00	Received	ASF	Donation	22-Jun		Anonymous	
	$ 100.00	Received	ASF	Donation	23-Jun		Anonymous	
	$ 55.00	Received	ASF	Donation	26-Jun		Kenneth Yeung	
	$ 100.00	Received	ASF	Donation	30-Jun		Amelia Tang	
Total (AUD)	$ 44,610.00							
Remaining (AUD)	$ 20,390.00							
Total (GBP)	£22,781.10							
Total (SAR)	SAR 110,386.70							

Highlighted how much left to raise

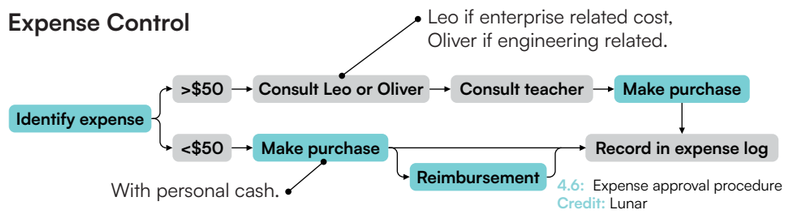


OUTFLOWS							
Seller	Details	Category	Actual cost (AUD)	Date	Status	Doc No	Team member
Objective 3D	3D parts	Car	$ 116.74	21/05/2024	Paid	INV-3037803	Liam
Objective 3D	3D parts	Car	$ 81.40	30/05/2024	Paid	INV-3037887	Liam
AMF Magnets	Magnets	Car	$ 11.60	9/07/2024	Paid	Order 100008104058	Liam
Bunnings Warehouse	Spray paints	Car	$ 182.88	15/07/2024	Paid	*Physical receipt	Thivi
S-Trend	Secondary uniforms	Uniform	$ 261.90	18/08/2024	Paid	11603	Leo
F1 in Schools Ltd.	Registration	Travel	$ 14,828.38	10/08/2024	Paid	11860	Thivi
Qatar Airways	Flights	Travel	$ 26,400.00	30/08/2024	Paid		Thivi
Novotel Darmnam	Hotel	Travel	$ 10,888.52	30/08/2024	Paid		Thivi
Paper Tiger	Pit Display boxes	Pit Display	$ 1,099.65	6/09/2024	Paid	295	Thivi
Waebuy Pty Ltd	18V Charger	Pit Display	$ 180.00	14/09/2024	Paid	*Physical receipt	Thivi
Officeworks	Mailing box	Pit Display	$ 23.34	15/09/2024	Paid	*Physical receipt	Thivi
Supercheap auto	Paints	Car	$ 66.77	15/09/2024	Paid	*Physical receipt	Thivi
Mantra Bearings & Industrial supplies	Ball bearings x12	Car	$ 69.40	25/09/2024	Paid	12784	Thivi
Total (AUD)			$ 56,250.38				
Total (GBP)			£ 29,177.69				
Total (SAR)			SAR 136,468.44				

AUD, GBP, SAR currency conversion

4.5: Income log extract (top) and expense log extract (bottom)
Credit: Lunar

Expense Control



4.6: Expense approval procedure
Credit: Lunar

Budget Tracking & Changes

Updated flight costs

Travel budget reduced as project went over budget

	APR ($)	MAY ($)	JUN ($)	JUL ($)	AUG ($)	SEP ($)	OCT ($)
Travel	59,000	↓47,000	47,000	47,000	↑58,840	7,000	↓3,000
Cars	4,000	↓4,000	3901	3804	2901	2111	100
Pit Display	5,000	5,000	5,000	↓3,000	3,000	1,900	↓600
Uniform	2,000	↓1,000	1,000	1,000	638	↑2800	0
Printing	2,000	↓1,000	960	960	766	766	↓0
Contingency + Other	8,000	↓7,000	7,000	7,000	7,000	7,000	↓178
Operating Capital:	80,000	↓65,000	64,861	62,764	↑74,244	↑-4,910	↓3,878

The budget we calculated at initiation was seen to be unrealistic, thus it was changed.

Simple PD design chosen for ease of use over aesthetics, also being cheaper.

Changes with no colour are when expenses are made, but the allocated budget does not change.

4.7: Total category budget changes as expenses and adjustments are made
Credit: Lunar

KEY:
↓ = decrease in total allocation
↑ = increase in total allocation

Final Evaluation (as at 31st October)

Expected spending: $65,000
Actual spending: $58,941.68
Variance: $6,058.32

Raised Cash: $62,820
Final spending: $58,941.68
Remaining contingency: $3,878.32

To spend on food/incidental expenses

ROLES, RESPONSIBILITIES & COMMUNICATIONS

TEAM ROLES

Leo Jansen | Team Manager



Skills: Team management, Project management, Graphic design
Knowledge: Agile methodology, Adobe design suite, Meta business suite
Experience: 2 years as Team Manager

Oliver Chernikeeff | Engineering Director



Skills: Aerodynamics, Kinematics, CAD software
Knowledge: Autodesk suite, Ansys electronics suite, Aerodynamic principles
Experience: 2 years as Design Engineer

Carey Tong | Enterprise Director



Skills: Marketing strategies, Sponsorship, Project management
Knowledge: Brand positioning, Meta business suite, Aerodynamic principles
Experience: 3 years as Team & Logistics Manager

Max Zylberman | Digital Engineer



Skills: Aerodynamics, Kinematics, CAD software
Knowledge: Autodesk suite, Ansys electronics suite, Adobe design suite
Experience: 1 year as Enterprise Director (now switching to better utilise skills)

Liam Jones | Manufacturing Engineer



Skills: CAM software, CAD software, Manufacturing techniques
Knowledge: Material properties, Autodesk suite, Quickcam pro
Experience: 2 years as Manufacturing Engineer

Kenneth Yeung | Logistics Manager



Skills: Public speaking, Enterprise, Marketing
Experience: 1 competition as Enterprise Director

Kenneth was added near the end of our project when it was unclear if Carey was able to attend the World Finals as a backup, and to help with final logistics.

Role Descriptions

Team Manager

Manages the project strategy team members and daily operations. Graphic Design of visual elements is a secondary role to this. Point of contact between team and ICC.
Key responsibilities: Scope snapshot, PM portfolio, Graphic design.

Engineering Director

Devises strategy for the engineering of the car, and participates in the modelling & testing of CAD prototypes and iterations.
Key responsibilities: CAD prototyping, Car testing, Engineering portfolio.

Enterprise Director

Develops strategy for the team's marketing activities, sponsorship acquisition and sustainability initiatives.
Key responsibilities: Marketing, Sponsorship, Enterprise portfolio.

Digital Engineer

Virtually tests, renders and evaluates CAD prototypes. Development of the pit display is secondary to this.
Key responsibilities: Car testing, Renders, Pit display.

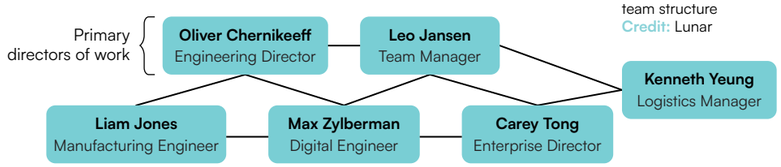
Manufacturing Engineer

Operating CAM software and hand-finishing cars, prototypes and testing materials.
Key responsibilities: CAM, Hand-finishing, Engineering portfolio.

Logistics Manager

Manages final task and packing logistics and reserve for Enterprise Director.
Key responsibilities: Project management logitstics, Pit display, Enterprise portfolio.

TEAM STRUCTURE



5.1: Our World Finals team structure
Credit: Lunar

Co-leadership Approach

Leo directed many of the project's general activities mostly in enterprise, and Oliver directed all of the engineering activities, using their collective skills to manage the project rather than one primary team leader. We also avoided a strict hierarchy, because we ensured our close connectedness rather than hierarchal respect. This was developed during our re-evaluation when our initial team structure including a predominant hierarchy proved ineffective, as it limited communication between lower 'ranking' members.

RACI MATRIX

Active Input: R = Responsible; A = Accountable
Passive Input: C = Consulted; I = Informed

	LEO	OLIVER	CAREY	MAX	LIAM	KENNETH
SCRUTINEERING						
Engineering drawings	I	A		R	C	
Rendering	I	I		R	C	
Quality of finish & assembly		C			R	
DESIGN & ENGINEERING						
Design Concepts	I	R		I	I	
3D modelling		R			I	
Computer aided analysis	I	A		R		
Use of CAM/CNC	I	C		I	R	
Other manufacturing & assembly	I	A			R	
Research & development	I	R		A	R	
Testing	I	R	I	A	R	
Design process evaluation		R	C	A	A	
Document presentation	R	A	A	C	C	
PROJECT MANAGEMENT						
Initiation process	R	I	C	I	I	C
Project schedule	R	I	C	I	I	I
Budget & resource management	R	I	C	I	I	I
Roles & responsibilities	R	I	C	I	I	C
Team/stakeholder communications	R	I	C	I	I	I
Risk management	R	I	C	I	I	I
Monitoring & controlling	R	I	C	I	I	A
ENTERPRISE						
Marketing	C		R			A
Sponsorship	A	A	R	A	A	A
Digital media	R		R		I	
Sustainability	A	A	R	A	A	A
Document presentation	R		A	C		
Overall team identity	R	I	C	I	I	
Pit display design process	C			R		
Pit display content	R	I	I	A	I	
VERBAL PRESENTATION						
Visuals	C	C	C	R	C	C
Team contribution	R	A	A	A	A	A
Engagement	R	R	R	R	R	R
Concept clarification	R	R	R	R	R	R
Time/presentation	R	A	A	A	A	A
Innovation	R	R	R	R	R	R
Collaboration	R	R	R	R	R	R
F1 in Schools learning experiences	R	R	R	R	R	R

Using the 5 scorecards, we delegated certain criterion to each team member rather than specific deliverables. This scorecard centred approach determined how much effort we apply to each, ensuring we are allocating our work efficiently to the points awarded.

5.2: Our full World Finals RACI matrix
Credit: Lunar

Sponsorship

Because sponsorship is attained mostly through personal contacts, every team member is accountable for finding people through their own network to achieve the most success, being directed by Carey.

Sustainability

Everyone is accountable for their sustainable considerations (see Enterprise Portfolio).

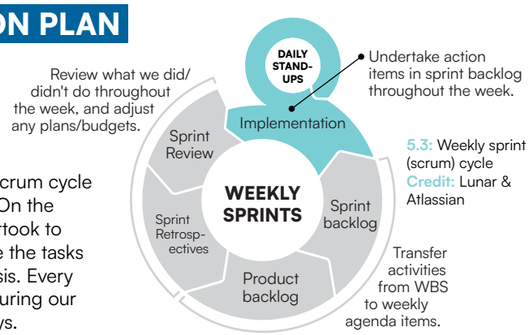
Verbal presentation

Because the verbal presentation is delivered by the team, everyone is responsible for speaking their parts. Leo was responsible for keeping track of time and managing the script, and Max was in charge of the visuals.

COMMUNICATION PLAN

Sprint Cycle

To break down our project into more manageable phases of work, we followed an adapted traditional agile scrum cycle to undertake weekly sprints. On the right are the stages we undertook to communicate and coordinate the tasks we complete on a weekly basis. Every stage in grey is undertaken during our weekly meetings on Thursdays.



5.3: Weekly sprint (scrum) cycle
Credit: Lunar & Atlassian

Communication Tools

PLATFORM	CONTEXT	FREQUENCY
Microsoft Teams	Formal text & video digital conversations, file storing, app integration (flexible customization). External stakeholder meetings are also possible here.	Daily
WhatsApp	Casual text & video conversations for all internal messages.	Daily
Jira Software	Task management with agile board for tracking progress, sprint planning & interactive Gantt chart.	Weekly
Microsoft OneNote	Collaborative writing of all documents.	Daily
Adobe Illustrator (cloud)	Collaborative illustrating & document formatting.	Daily
Fusion (team)	CAD modelling of car & pit display	Daily
Email	External stakeholder communication & reporting to school teachers.	≈4 times a week
Excel	Master finances & stakeholder spreadsheet (later replaced Jira Software).	≈4 times a week

5.4: Breakdown of our communication tools for team members and other stakeholders
Credit: Lunar

STAKEHOLDER MANAGEMENT

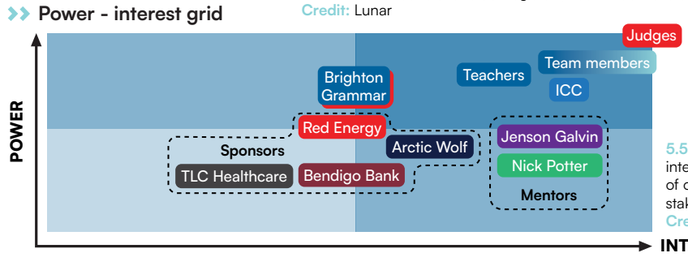
Stakeholder Register

Classification

STAKEHOLDER	INSTITUTION		COLLABORATION
Team members	Lunar	Internal	Project initiating, planning, executing, monitoring
Judges	F1 in Schools Ltd	Key stakeholder	Competition interviewers, points evaluators (KPI)
School	Brighton Grammar	Internal	Competition registration, assistance with deliverables
Teachers in Charge	Brighton Grammar	Internal	Assistance with deliverables, supervising
Jenny Duff	Red Energy	Title sponsor	Funding, pit display energy
Simon Walker	Arctic Wolf	Principal partner	Funding, external marketing
Michael Myers	REA Foundation	ICC	Co-ordination of team & World Finals event
Mary Munafo	Bendigo Bank	Tier 3 sponsor	Funding, team uniform, external marketing

5.5: Extract of our stakeholder register (most influential stakeholders)
Credit: Lunar

Power - interest grid



5.5: Power - interest grid of our main stakeholders
Credit: Lunar

RISK MANAGEMENT

Stakeholder Communication Plan

STAKEHOLDER			COMMUNICATION	FREQUENCY
Brighton Grammar School	5	3	▪ Finance information ▪ Status reports	Fortnightly
Supervising teachers	5	5	▪ One on one mentoring ▪ Finances information ▪ Physical meetings ▪ Project data	3 times a week
Ken Lau (Qatar Airways)	5	1	▪ Monthly emails ▪ Arranged ROI tailoring meetings ▪ Finance information	Monthly
Simon Walker (Arctic Wolf)	4	3	▪ Monthly emails ▪ Arranged ROI tailoring meetings	Monthly
Jenny Duff (Red Energy)	4	3	▪ Monthly emails ▪ Arranged ROI tailoring meetings	Monthly
Michael Myers (ICC)	4	3	▪ General emails	*Irregular
Nick Potter	3	2	▪ Physical feedback meetings	Monthly
Jenson Galvin	2	5	▪ Physical feedback meetings	Fortnightly
Mary Munafo (Bendigo Bank)	2	4	▪ Community connections meetings ▪ Monthly emails	2 times a month

6.1: Communication plan for each stakeholder identified in our stakeholder register extract
Credit: Lunar

RISK IDENTIFICATION

Risk Register

Impact · Probability · Risk Appetite

RISK				AREA OF IMPACT	URGENCY	OWNER
CAD DESIGN						
Key software license expiry	6	2	12	Timing, Quality	Medium	Max
Key hardware failure	7	2	14	Resources, Timing, Quality	Medium	Max
Prototype deadlines not met	7	7	49	Timing	Medium	Oliver
Final design freeze not met	7	6	42	Timing	High	Oliver
Doesn't comply with regulations	8	5	40	Quality, Scope	Medium	Oliver
Loss of CAD files	5	2	10	Timing	Low	Oliver
Poor design quality	4	9	36	Resources, Timing, Quality	Medium	Oliver
CAR MANUFACTURING						
Shortage of resources	6	5	30	Resources, Quality	Medium	Liam
Equipment breakdowns	8	2	16	Scope, Timing, Quality	Medium	Liam
Physical injury	5	2	10	Timing	Low	Liam
Supplier delays	6	2	12	Timing	Low	Liam
Manufacturing errors/breakages	4	9	36	Timing, Quality, Resources	Medium	Liam
Not met deadline	10	8	80	Timing	Medium	Liam
Car breakage during transit	9	5	45	Quality, Points	High	Liam
Car doesn't hit our key results	5	4	20	Scope	Low	Liam
Car breaks regulation before work freeze	4	9	36	Timing, Resources	Medium	Liam
Car breaks regulation at competition	10	7	70	Points	High	Liam
PIT DISPLAY						
Design delays	6	7	42	Timing	Medium	Max
Supplier delays	7	3	21	Timing	Low	Max
Breaks during staging	5	6	30	Resources, Quality, Timing	Medium	Max
Breaks at competition	10	6	60	Quality	High	Everyone
OTHER						
Missed submission	10	8	80	Points	Irreversible	Leo
Lack of financial backing	10	8	80	Resources, Quality, Scope	Irreversible	Carey
Burnout	8	9	72	Timing, Quality	Ambiguous	Leo
Team member absence	5	3	15	Timing, Quality	Uncertain	Everyone
Scope creep	5	7	35	Scope	Ambiguous	Leo
Regulation changes	3	5	35	Scope, Timing	Uncertain	Everyone
Inadequate initial planning	8	5	40	Scope, Timing	Ambiguous	Leo

These 2 risks were added after we actually experienced them.

6.2: Our full World Finals risk register
Credit: Lunar

Risk Appetite

Probability % \ Impact %	0	2	4	6	8	10
10	0	20	40	60	80	100
8	0	16	32	48	64	80
6	0	12	24	36	48	60
4	0	8	16	24	32	40
2	0	4	8	12	16	20
0	0	0	0	0	0	0

Impact × Probability = Risk Appetite (%)

6.3: Our risk appetite matrix
Credit: Lunar

RESPONSE PLANNING

Risk Appetite Statement

RISK LEVEL	PHILOSOPHY	TOLERANCE FOR UNCERTAINTY	CHOICE when faced with multiple options	TRADE-OFF against objectives
Low 0-20%	We will take justified risks	Fully anticipated	Highest return, accept the possibility of failure	Willing
Medium 21-49%	Preference for safe delivery	We will accept if limited + outweighed by benefits	Will accept if limited and outweighed by benefits	Prefer to avoid
Critical 50-100%	Avoidance is a core objective	Very low	Always choose the lowest risk option	Never

6.5: Our risk appetite statement - how we will treat each risk appetite level
Credit: Lunar

Risk owners and urgency

Each risk owner is responsible for avoiding and mitigating the impact of their risks, guided by the above statement, which is taken into effect depending on the urgency of the control (largely decided by the impact):

- **Low:** Can neglect if not impacting work
- **Medium:** Begin to control, but don't neglect other deliverables
- **High:** Immediately implement control, neglect current action items until solved

These control strategies are up to the risk owner to decide, which is discussed in our weekly meetings and stand-ups.

Critical Risk Mitigation

Missed submission deadline

Inherent risk: Rushing to finish portfolios last minute could lead to less time for preparing verbals, risking point loss from inadequate preparation. Additionally, late submission will result in significant point penalties.
Control measure: Introducing the schedule separation of work and refinement to ensure we complete our portfolios long before the deadline, so we can spend the rest of the time to refine and respond to feedback.
Impact of control: 80 → 35 { Impact: 10 → 7 ; Probability: 8 → 5

Residual risk: By completing portfolios this early could mean insufficient time was spent on them, leading to significant changes later that may impact remaining tasks.

Lack of financial backing

Inherent risk: Insufficient cash to fund the project, reducing quality of deliverables and the inability to cover $56,400 that was essential to the project (see below).
Control measure: Each member initially contributed $5,000 of personal capital, meaning a minimum of $31,400 needed to be raised to compete. This cash was refunded once we secured enough to compete comfortably.
Impact of control: 80 → 35 { Impact: 10 → 7 ; Probability: 8 → 5

Residual risk: It is still always the case that our budget could be exceeded, leading to reduced quality of deliverables. This is why our expense control (figure 4.6) is closely managed.

Travel	$47,000
Cars	$2,000
Pit Display	$1,500
Uniform	$350
Printing	$550
Contingency & Other	$5,000
Total:	**$56,400**

Pit display breakage

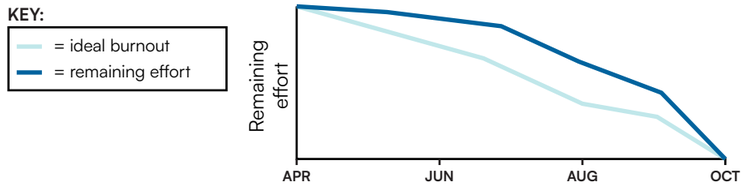
Inherent risk: Pit display not arrived or broken, leading to significant point loss.
Control measure: Bringing extra roll-up banners in our carry-on luggage to use if our full pit display doesn't arrive.
Impact of control: 49 → 35 { Impact: 7 → 5 ; Probability: 7 → 7

Residual risk: Just having banners as our pit display will still lead to points loss, but better than no pit display.

Burnout

Inherent risk: Decreased productivity and quality of work, leading to missed deadlines and increased stress.
Control measure: Introducing evenly distributed milestones and scheduled breaks from the project upon the completion of these milestone. This later became additional rest periods after burnout was experienced near the end of our project. Our team manager was responsible for balancing these rest/work periods when needed.
Impact of control: 90 → 36 { Impact: 9 → 6 ; Probability: 10 → 6

Residual risk: Even with scheduled breaks and milestones, some team members may still face fatigue or burnout during peak periods, affecting productivity. Breaks from the project also mean no work being completed, delaying completion of work.

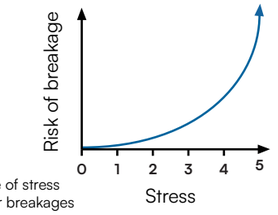


6.8: Burnout chart illustrating the effort from our team (approximate)
Credit: Lunar

OTHER RISK CHANGES

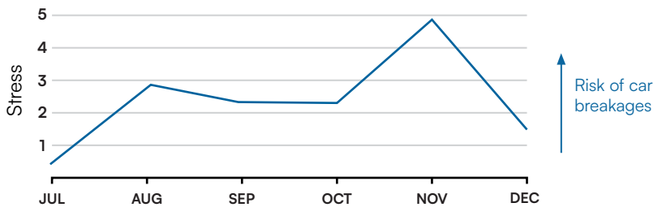
Car Breakages

It's proven that as stress levels increase (as our deadlines approach) mistakes in manufacturing are exponentially more likely. This is what we experienced firsthand on some occasions, illustrated in the graph on the right, prompting the team welfare assessment in our status reports (introduced in July).



7.1: Influence of stress on risk of car breakages
Credit: Lunar

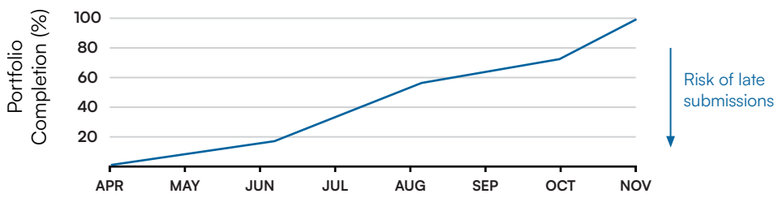
MONITORING & CONTROLLING



7.1: Change in welfare levels as deadlines approach
Credit: Lunar

Late Submissions

The risk of submitting our portfolios late decreases as we complete them, thus being mitigated by setting and hitting set deadlines.



7.2: Risk of submitting our digital elements late in relation to their completion
Credit: Lunar

EVALUATING

Meeting Minutes & Status Reports

Every meeting agenda for our weekly progress team meetings was written a few days in advance for other team members to input agenda items. The minutes from these meetings formed the basis for our monthly status reports.

Our status reports did not need to be published more frequently than monthly, as our weekly meeting minutes also provided a detailed status update on a weekly basis, thus our status reports acted more as a reflection of progress and more detailed monitoring of the project, which may uncover issues that get overlooked during weekly meetings.

These were always done by Leo, the Team Manager, and published to all team members, supervising teachers and relevant sponsors and other stakeholders to inform them of our progress.

Pit lane walk: What we call a period of 60 seconds for each team member to recap their previous week's tasks to everyone.

Row above: Agenda item.
Row below: Minutes about the above agenda item.

Budget/sponsors: A recap of our financial situation.

Schedule: Where we discuss any upcoming deadlines.

Issues and scope creep: An opportunity for any team member to identify any current issues and/or scope creep.

Action items 1-4 are repeated every week.

Other agenda items: These items are prepared a day in advance by the Team Manager, to plan discussions about specific deliverables or upcoming events.

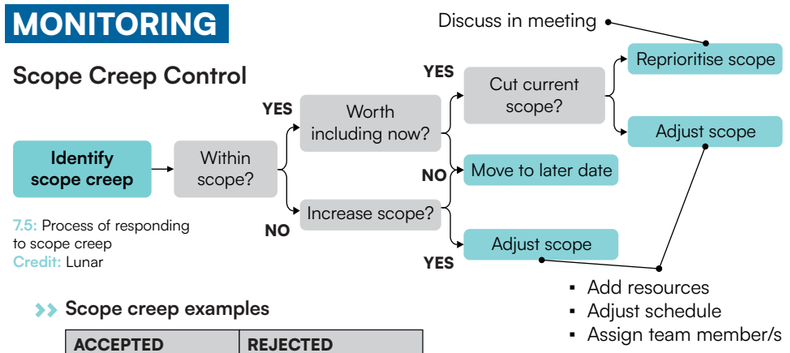
AOB: Minutes for any other discussions that happen.

Next action items: Assigned tasks for each team member to complete by the meeting next week (what is placed in "Action items from last week" at the top of next meeting's agenda.

Highlighted sections: Important notes.

MONITORING

Scope Creep Control



7.5: Process of responding to scope creep
Credit: Lunar

Scope creep examples

ACCEPTED	REJECTED
Create custom cars for sponsors	Team jackets
Add team member (Kenneth)	More finances allocated to Pit display
Gather resources for future competitions	Increased total budget

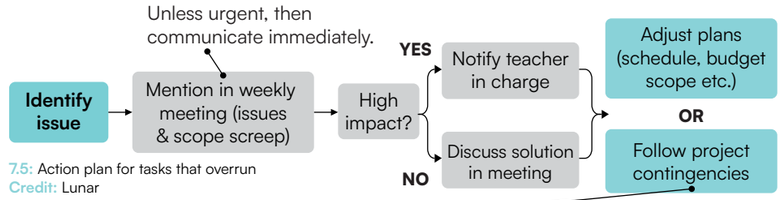
Although cool, they were too expensive.

As discussed in our project scope, we could not afford to dedicate too many resources to our pit display due to the little points awarded for it.

We decided to spend extra cash on longer term assets after we agreed we would compete in F1 in Schools next year.

7.6: Examples of scope creep identifed throughout our project
Credit: Lunar

Action Plan for Tasks that Overrun



7.5: Action plan for tasks that overrun
Credit: Lunar

Budget: 12% of full budget ($7,000).
Schedule: Schedule beyond Work Freeze (17 August) or other later date.
Supplier: See figure 4.3.

WEEKLY MEETING MINUTES

Date: Thursday 4/7/24 **Location:** F1 workshop
Time: Lunchtime 1:45pm **Team members:** All (Carey join virtually)

ACTION ITEMS FROM LAST WEEK:

Leo = Go through sustainability plan with Carey, Layout rest of Enterprise Portfolio.
Oliver = Finish car — send to Brett. Continue car development pages.
Carey = Finish Marketing Strategy & go through sustainability with Leo.
Max = Figure out box layout, finish what is completable of pit display page, scrutineer car, finish wheel page.
Liam = Finish car weight calculations, Manufacture refinement car, finish CAM/CNC page.

AGENDA ITEMS:

Item	Topic	Presenter
1	**Pit lane walk**	Everyone
	Leo = Briefly gone through Sustainability (still to work on), almost finished Enterprise Portfolio layout, done PD banner graphics. **Oliver** = Fixing up car (will finish Friday), then send to Brett. Not finished portfolios (will after car) **Carey** = Not yet finished Marketing strategy, but have plan for sustainability, also through Engineering portfolio and fixed up the layout **Max** = Finished PD structure and PD page (to be refined), yet to do car development pages and will scrutineer car when Oli is done car. **Liam** = Done weight calculations (except for what is dependent on Oli), doing refinement car, done Manufacturing page.	
2	**Budget/Sponsors**	Leo
	• Now have negative cashflow • Pending sponsors: SSS ($5,000), Travis, Lou ($10,000) • Upcoming expenses: Pit display structure ($1,000)	
3	**Schedule**	Leo
	Adjusted portfolio deadline (Thursday 27 September)	
4	**Issues & Scope Creep**	
	• *Engineering portfolio taking too long — need to sort out! • No scope creep identified	
5	**Pit Display Mock-ups**	Max
	• 2 Options (banner vs no banner) • Still figuring out what is to go on the boxes if no banners	

*Fill in during meeting

ANY OTHER BUSINESS (AOB):

*None

*Fill in during meeting

NEXT ACTION ITEMS:

Leo = Sort out enterprise (sustainability & other graphics), order uniforms, PD designs for boxes.
Oliver = Finish car (Friday) — portfolio layout rest of Engineering portfolio
Carey = Help layout engineering, Go through Digital media/sponsorship
Max = Scrutineer car (Friday), finalise pit display table, start filling in engineering, fix up PD file
Liam = Finish CAM/CNC page, finished manufacturing refinement car (ready before USA trip).

7.3: Example of one of our meeting agendas (4/7/24)
Credit: Lunar

STATUS REPORT

Project: Lunar F1 in Schools 2024 World Finals **Reporter:** Leo Jansen
Date: 30 June 2024 **Period:** 1 June - 1 July 2024

PROJECT SUMMARY:

GENERAL STATUS	Moderately behind schedule		
CURRENT ACTION ITEMS	**AHEAD ON:** • Pit display design • Car design	**ON TOP OF:** • PM portfolio • Enterprise portfolio	**BEHIND ON:** • Engineering portfolio • Interim uniforms • Social media
DELIVERABLES	**COMPLETED** • Logo & branding • Prospectus • Polo designs • Pit display structure design	**INCOMPLETE** • Verbal presentation script • Pit display • Polo uniform order • Other uniform	
ENGINEERING SUMMARY	• Currently iterating front wing - final part of the car. Will meet the design freeze of 13/6. • Track testing revealed X toe is fastest - yet to validate with ramp. • Yet to test bearings, paint, machining plans.		
PORTFOLIO SUMMARY	**PROJECT MANAGEMENT:** • **Completed:** Initiation, Schedule, Roles & Communication • **In progress:** Risk Management • **Incomplete:** Monitoring, Budget & Resources, Graphics **ENTERPRISE:** • **Completed:** Digital media, Sustainability • **In progress:** Branding, Marketing, • **Incomplete:** Trade Display, Graphics **DESIGN & ENGINEERING:** • **Completed:** Digital media, Sustainability • **In progress:** Branding, Marketing, • **Incomplete:** Trade Display, Graphics		
BUDGET	**Revenue:** $36,510.00 **Expenses:** $198.14 **Varience:** $36311.86	**Upcoming Revenue:** $8,000 (July - Arctic Wolf sponsor) **Upcoming Expenses:** Uniform order, magnets, Spray paint, 3D printing	

AREAS FOR CONCERN:

ISSUES & ACTION PLANS	**ISSUES:** • Carey away - unsure of return • Engineering portfolio - nearing deadline with little progress.	**ACTION PLAN** • Deligate Carey's current tasks to Leo and Max • Gathering support from supervising teachers to hold team accountable for work
RISKS	• Leo away 16-20 July • Max away 10-17 July • Sponsorship funding - still $20,490 remaining - will need to follow up prospective sponsors and seek further sponsorship.	
SCOPE CREEP	• Potential scope creep in manufacturing of other cars for sponsors - may accept • Projected cost for pit display ($7,000)	

CONCLUSION:

- Pit display order pushed back to confirm sponsor logos on banners (date to be confirmed).
- Interim uniform not yet ordered (Non-critical delay)
- **4 July:** Project Management Portfolio set to be completed
- **13 July:** Car design freeze
- **17 July:** Majority of portfolios set to be completed

TEAM WELFARE	7/10

Documenting tasks signed off.

We decided to separate the tasks associated with Portfolios and Engineering since they are worth the most amount of points, so we wanted a more detailed description of these tasks to ensure they are effectively tracked.

Action plan identified for tasks that overrun (using figure 7.5 as a guide).

Team Welfare: General score of how the team feels about the project (higher number = more positive).

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Thank you for reading Lunar's 2024 World Finals Project Management Portfolio!